NYC Airbnb Price Prediction

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Dataset"New York City Airbnb Open Data" by AB NYC(from kaggle)

Data Loading & Inspection

→Load the New York City Airbnb dataset into a structured dataframe to analyze its overall shape, dimensions, and column variables.
→Inspect the first few rows of data to observe how values are recorded across different attributes.

Attribute Categorization:

Numerical Columns: Identify discrete and continuous quantitative variables such as price, minimum_nights, number_of_reviews, reviews_per_month, calculated_host_listings_count, availability_365, latitude, and longitude.
Categorical Columns: Identify nominal text variables such as neighbourhood_group, neighbourhood, room_type, name, host_name, and last_review.

Missing Value:

Perform a comprehensive check across all columns to detect missing or null entries.
Calculate the total count and percentage of missing values for each column to determine necessary data imputation or removal strategies in subsequent steps.

Data Cleaning & Handling Missing Values

Handling the last_review Column: Look closely at the last_review column. Since it is a date string containing over 10,000 missing values—and standard linear regression models cannot process raw dates without complex feature engineering decide to drop the column entirely.
Imputing Missing Values in reviews_per_month:
Investigate the missing values within the reviews_per_month column.
Realize that these values are missing because those specific listings have received zero reviews.
Fill these missing data points with a logical replacement value of 0.
Dropping Irrelevant Identifiers:
Remove unique text identifiers and columns that cannot help a mathematical model learn price patterns

Outlier Identification & Treatment

Visual Detection of Outliers: Create visual plots such as boxplots and histograms for variables like price and minimum_nights to easily detect extreme values and abnormalities.

Capping and Trimming Strategy: Apply the 95th percentile rule to filter out extreme luxury penthouses (such as $10,000 per night listings). Filter out abnormal long-term rental rules (such as minimum stays of 365+ nights) so that the regression model remains focused strictly on the typical consumer market rather than market distortions.

Feature Selection & Categorical Encoding

Addressing High Cardinality: Examine the categorical columns: the neighbourhood column contains over 200 unique values, and the neighbourhood_group column contains 5 broad values (boroughs).
Understand why keeping the 200+ unique values would break the model's simplicity—a problem known as the "Curse of Dimensionality".
Drop the hyper-specific neighbourhood column to keep the model clean and manageable.

Categorical Encoding (One-Hot Encoding): Convert the remaining categorical columns (neighbourhood_group and room_type) into a numerical format using One-Hot Encoding.Ensure that the first column is dropped (drop_first=True) to avoid multicollinearity and the "Dummy Variable Trap".

Feature-Target Separation & Validation Split

Feature and Target Separation: Separate the cleaned dataset into independent feature variables (X) and the target variable (y), which represents the listing price we want to predict.

Training and Testing Data Split: Split the data into an 80% Training Set to teach the model and a 20% Testing Set to evaluate it.
Ensure that a specific reproducibility seed (random_state) is used during the split so that the results can be consistently verified.

Model Training & Evaluation

Model Initialization & Fitting: Import, initialize, and fit a standard Linear Regression model using the 80% training data.

Generating Price Predictions: Use the trained model to generate price predictions on the hidden 20% testing dataset

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Residual Analysis & Interpretation

Actual vs. Predicted Plot: Plot a scatter plot of actual prices against predicted prices. Draw a 45-degree diagonal line to show what a "perfect" prediction would look like.

Residual Plotting: Plot a Residual Plot (errors versus predicted values) or an error distribution histogram to check the distribution of model mistakes.

Final Project Question Answer

Question: Does the model perform better on cheap listings or expensive listings, and why?

Answer Summary: The model performs better on cheap-to-mid-range listings because they follow stable, predictable patterns based on features like room type and borough. Expensive listings have high variance and are harder to predict accurately due to uncaptured luxury amenities or qualitative factors.